

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building is 45° . The height above his eye level is:
 - (A) (A) 30 m
 - (B) (B) 28.5 m
 - (C) (C) 31.5 m
 - (D) (D) 1.5 m
2. The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60° . The height of the tower is not required, but if we consider a simpler case where the shadow is $h\sqrt{3}$ when angle is 30° , what is the height h if the shadow is 30 m?
 - (A) (A) $30\sqrt{3}$ m
 - (B) (B) 30 m
 - (C) (C) $10\sqrt{3}$ m
 - (D) (D) 10 m
3. If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then the length of OP is:
 - (A) (A) r
 - (B) (B) $2r$
 - (C) (C) $r\sqrt{3}$
 - (D) (D) $r/2$
4. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:
 - (A) (A) 10 cm
 - (B) (B) 8 cm

- (C) (C) 12 cm
(D) (D) 7 cm
5. The number of tangents that can be drawn from a point lying inside a circle is:
- (A) (A) 0
(B) (B) 1
(C) (C) 2
(D) (D) Infinite
6. If two tangents inclined at an angle of 80° are drawn to a circle of radius 3 cm, then the angle between the two radii at the center is:
- (A) (A) 80°
(B) (B) 90°
(C) (C) 100°
(D) (D) 120°
7. The length of a tangent from a point A at a distance of 10 cm from the center of the circle is 8 cm. The radius of the circle is:
- (A) (A) 4 cm
(B) (B) 6 cm
(C) (C) 7 cm
(D) (D) 9 cm
8. Assertion (A): The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 45° . The height of the tower is 30 m.
Reason (R): In a right-angled triangle, $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the Sun is 45° . Reason (R): The angle of elevation is the angle formed by the line of sight with the horizontal when the object is above the horizontal level.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

- (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): The length of a kite string making an angle of 30° with the ground to reach a height of 50 m is 100 m. Reason (R): $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
11. Assertion (A): The angle of depression from a point 10 m above sea level to a ship at sea is 30° . The distance of the ship from the base of the point is $10\sqrt{3}$ m. Reason (R): Angle of depression is equal to the angle of elevation by alternate interior angles.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
12. Assertion (A): If a ladder 10 m long is placed against a wall such that it makes an angle of 60° with the ground, then the foot of the ladder is 5 m away from the wall. Reason (R): $\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
13. Assertion (A): The length of a tangent drawn from an external point to a circle is equal to the length of another tangent drawn from the same point. Reason (R): The lengths of tangents drawn from an external point to a circle are equal.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

14. Assertion (A): The tangent at any point of a circle is perpendicular to the radius through the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
15. Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A point inside a circle lies in the interior, and any line passing through it will be a secant, not a tangent.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
16. Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then $OP = 2r$. Reason (R): The line segment from the center to the external point bisects the angle between the tangents.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
17. Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the center of the circle is 4 cm. The radius of the circle is 3 cm. Reason (R): In a right triangle formed by radius, tangent, and hypotenuse (distance to center), $r^2 + t^2 = d^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

1. A ladder 15 m long makes an angle of 60° with the wall. Find the distance of the foot of the ladder from the wall.
2. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30° . Find the height of the tower.
3. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string.
4. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° . If the distance between the building and the tower is $7\sqrt{3}$ m, find the height of the tower.
5. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked.
6. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. Find the length of PQ .
7. If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then find $\angle POA$.
8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
9. A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .
10. If a point P is 26 cm away from the center O of a circle and the length of the tangent PT drawn from P to the circle is 10 cm, find the radius of the circle.

Section C: Short Answer (3 marks each)

1. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 m. At a point on the plane, the angle of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. Find the height of the tower.
2. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.
3. The angle of elevation of the top of a lighthouse from a boat is 30° . If the boat moves 100 m towards the lighthouse, the angle of elevation becomes 45° . Find the height of the lighthouse.
4. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string.

5. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.
6. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
7. From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.
8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
9. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , then find $\angle POA$.

Section D: Long Answer (4-5 marks each)

1. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h . At a point on the plane, the angles of elevation of the bottom and the top of the flagstaff are α and β respectively. Prove that the height of the tower is $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$.
2. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.
3. The angle of elevation of a cloud from a point 60 m above a lake is 30° and the angle of depression of the reflection of the cloud in the lake is 60° . Find the height of the cloud from the surface of the lake.
4. Two poles of equal heights are standing opposite each other side by side and opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles.